Impact of Within-Content Engagement Information: Evidence from YouTube's Engagement Graphs

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Abstract

Digital platforms strategically choose what information to provide to their users. Using a uniquely constructed dataset of over 121,000 YouTube videos collected in 2023, we examine the causal impact of engagement graphs—a novel form of within-content engagement information that displays moment-by-moment user engagement levels throughout a video. Our identification strategy exploits YouTube's batch processing approach to engagement graph activation: we reverse-engineer the platform's criteria, revealing that activation depends on a common view threshold and an eligibility date determined solely by upload timing, with batch processing creating quasi-random variation in treatment timing across otherwise similar videos. We leverage this variation to implement a difference-in-differences strategy with matched samples. Our analysis finds that engagement graphs increase content viewing, liking, and commenting in post-treatment periods, while leaving comment sentiment and depth unchanged. To deepen our understanding of these effects, we use large language models to construct transcript-based measures of each video's informational content and narrative structure, and estimate how these characteristics moderate the treatment effect. The graph's impact is concentrated among videos whose value is harder for viewers to assess as they watch—for example, videos that deliver their payoffs later and introduce new concepts more frequently—whereas narrative structure plays no moderating role. On the supply side, exploiting the feature's platform-wide rollout, we find no evidence that creators adjust their content production volume or characteristics, suggesting the feature's primary impact operates through the demand side. Our findings have important implications for platform design, content discovery, and understanding how granular engagement information shapes user behavior in digital environments.

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